Lab 1 — Use-Case Flow Specification

Problem Statement #12: Blood Bank Inventory & Emergency Donor Matcher

UC-01 — Process Emergency Blood Request

Primary Actor: Emergency Requester
Supporting Actors: Blood Bank Manager, Donor, SMS Gateway
Goal: Fulfil an emergency blood requirement using available inventory or, when inventory is insufficient, locate and notify compatible eligible donors.

Preconditions

1. The Emergency Requester is authenticated and authorized to submit an emergency request.
2. Blood bank inventory data is available to the system.
3. Registered donor records contain blood-group, eligibility, contact and location information.
4. The emergency notification service is operational.

Postconditions

Success: Required blood units are reserved from available inventory, or compatible eligible donors are notified and their responses are recorded.
Failure: The request remains pending/unfulfilled and no incompatible or expired blood unit is allocated.

Main Success Scenario

1. The Emergency Requester selects **Create Emergency Blood Request**.

2. The system displays the emergency request form.

3. The requester enters blood group, component type, number of units, hospital/location and urgency.

4. The system validates the entered information.

5. The system creates a unique emergency request ID.

6. The system searches real-time inventory for compatible and non-expired blood units.

7. The system identifies sufficient available units matching the request.

8. The Blood Bank Manager confirms allocation of the required units.

9. The system atomically marks selected units as allocated so they cannot be assigned to another request.

10. The system updates the request status to **Fulfilled**.

11. The requester receives confirmation that the required blood has been allocated.

12. The use case ends successfully.

Alternate Flow — A1: Insufficient Blood Inventory

A1.1 At Step 7, the system determines that sufficient compatible blood units are not available.

A1.2 The system initiates emergency donor matching.

A1.3 The system identifies registered donors with compatible blood groups who satisfy configured eligibility rules and are located within 10 km of the emergency location.

A1.4 The system sends an emergency SMS notification to matched donors through the SMS Gateway.

A1.5 A notified donor selects **Accept** or **Decline**.

A1.6 The system records the donor response against the corresponding emergency request.

A1.7 The Blood Bank Manager views responding donors and coordinates the required donation process.

A1.8 The request remains **Pending Donor Fulfilment** until sufficient blood is obtained and processed.

Business Rules

• Expired blood units shall never be considered available.

• A blood unit already allocated to another request shall not be allocated again.

• Only compatible and currently eligible donors shall be considered for emergency notification.

- Donor alerts shall be restricted to the configured 10 km emergency radius.
- Donor acceptance indicates willingness to donate; it does not itself make blood immediately available for transfusion.